
DATA SECURITY TECHNOLOGIES & STRATEGIES

Mbong Hudson
Infrastructure Engineer - CCNP, AWS, CISSP

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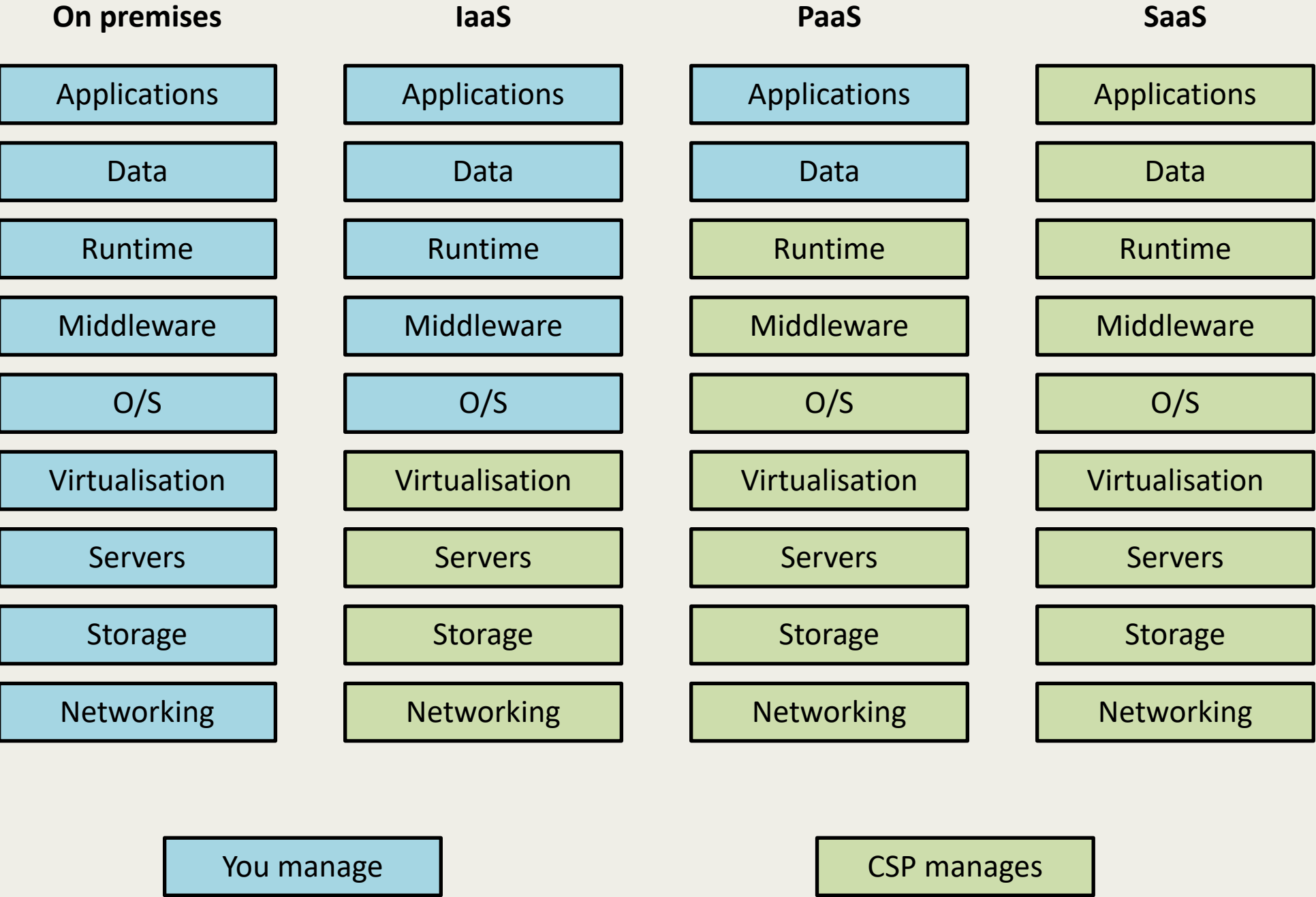
SCENARIO

Laura is reviewing a customer's file inside her organisation's customer relationship management tool and sees the customer's Social Security number listed as XXX-XX-8987. What sort of obfuscation technique has been used?

- A. Anonymisation
- B. Masking
- C. Randomisation
- D. Hashing

SHARED RESPONSIBILITY IN CLOUD SECURITY

- **Shared responsibility model:** Consumers secure their own data
- **Legal accountability:** Data owners are liable for breaches.
- **Collaboration:** Both cloud consumers and CSPs must secure hardware and facilities.



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ENCRYPTION & KEY MANAGEMENT

- **Encryption:** Transforms data into unreadable formats using keys and algorithms.
- **Kerckhoffs' Principle:** Security lies in protecting the key, not the system.
- **Key Lifecycle:** Creation, storage, use, sharing, archiving, and destruction.
- **FIPS 140-3 Standard:** Validates cryptographic modules.

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TYPES OF ENCRYPTION IN CLOUD ENVIRONMENTS

- **Storage-level encryption:** Protects physical storage (CSP-controlled keys).
- **Volume-level encryption:** Secures volumes for specific VMs (consumer-controlled keys).
- **Object-level encryption:** Consumer-encrypted objects before storage.
- **File-level encryption:** Application-based (e.g., Microsoft Word).
- **Application and database encryption:** Consumer-controlled keys, layered protection.

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HASHING: ENSURING DATA INTEGRITY

- **Hashing:** One-way encryption creating a unique fingerprint.
- **Uses:** Verifying data integrity, digital signatures, detecting system compromises.
- **Collision resistance:** Prevents two inputs from producing the same hash.
- **FIPS 180-4:** Guidance on Secure Hash Algorithm (SHA-3).

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PROTECTING DATA THROUGH OBFUSCATION

- **Substitution:** Replacing sensitive data with synthetic data.
- **Shuffling:** Rearranging data within a dataset.
- **Value variance:** Adjusting numerical values.
- **Deletion or redaction:** Nullifying or blacking out data.

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DATA MASKING

- Hides specific elements of sensitive data.
- Examples: "XXXXX1234" for bank account numbers.
- Prevents unintended disclosures.
- Works with structured data; challenges with unstructured data.

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DATA DE-IDENTIFICATION AND ANONYMISATION

- **De-identification:** Removes direct and indirect identifiers.
- **Anonymisation:** Permanent removal of identifiers.
- **Pseudonymisation:** Allows re-identification later.
- Combines techniques like shuffling, deletion, and substitution.

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KEY TAKEAWAYS

- Collaboration between cloud consumers and CSPs is essential.
- Encryption and key management are foundational to cloud security.
- Hashing, obfuscation, and masking ensure data integrity and limit exposure.
- De-identification techniques comply with privacy regulations.

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